E-Waste Management in India: A Secondary Data Analysis of Generation, Recycling, and Regulatory Challenges

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***Abstract*—The expansion of digital services, consumer electronics, and electrical equipment has made electronic waste (e-waste) an increasingly important environmental and resource-management issue in India. Safe management is challenging because electronic products contain both recoverable materials and components that can create environmental and occupational hazards when handled incorrectly. This paper examines reported national e-waste generation and recycling figures, reviews India's E-Waste (Management) Rules, 2022, and identifies implementation challenges that affect collection, traceability, and environmentally sound processing. The study uses secondary data from parliamentary and government publications, the Central Pollution Control Board (CPCB), and international e-waste reporting. For the five financial years from 2020–21 to 2024–25, the reported series shows that recorded generation rose from 1,346,496 metric tonnes in 2020–21 to 1,397,956 metric tonnes in 2024–25, with a peak of 1,609,117 metric tonnes in 2022–23. Reported recycling increased from 354,541 metric tonnes to 1,159,228 metric tonnes over the same period. The ratio of reported recycling to reported generation therefore increased from approximately 26.3% to 82.9%. These ratios describe the published national series and should not be interpreted as a complete measurement of all e-waste flows, especially where unreported or informal activity is not captured. The paper concludes that producer responsibility and registered recycling are important foundations, but they must be supported by accessible collection systems, transparent data, safer working practices, and greater emphasis on repair, reuse, and product-life extension.**

***Index Terms*—e-waste, India, Extended Producer Responsibility, recycling, circular economy, secondary data analysis**

I. Introduction

Electrical and electronic equipment supports communication, education, healthcare, commerce, and public services. Faster product replacement, increased device ownership, and the spread of digital infrastructure also create a growing stream of discarded equipment. Electronic waste, or e-waste, generally refers to discarded electrical and electronic equipment and its components. Examples include computers, mobile phones, printers, televisions, and household electrical appliances.

E-waste is not simply ordinary solid waste. Some equipment contains substances that require controlled handling, while many devices also contain materials with economic value, including iron, aluminium, copper, and selected precious metals. When products are dumped, burned, or dismantled using unsafe methods, useful resources may be lost and workers or nearby communities may be exposed to harmful substances. Conversely, suitable collection, refurbishment, and recycling can help recover materials and reduce demand for virgin resources.

The Global E-waste Monitor 2024 estimates that the world generated 62 million tonnes of e-waste in 2022, while only a minority was documented as formally collected and recycled in an environmentally sound manner [1]. The report estimates India's generation at approximately 4.1 million tonnes in 2022. This international estimate and India's administrative series are produced using different methods and scopes; they should therefore not be treated as interchangeable measurements.

India's regulatory framework includes the E-Waste (Management) Rules, 2022, notified by the Ministry of Environment, Forest and Climate Change (MoEFCC). The rules came into force on 1 April 2023 and established an updated Extended Producer Responsibility (EPR) framework supported by a centralized portal [2], [3]. The existence of a regulatory framework, however, does not alone establish how completely all discarded devices are collected or how effectively they are processed.

This paper reviews reported national figures for e-waste generation and recycling, explains the regulatory framework, and discusses the data and implementation challenges that remain relevant to sustainable management.

II. Research Problem, Objectives, and Questions

A. Research Problem

India's e-waste policy depends on the ability to bring discarded products into appropriate collection and treatment channels and to document their movement reliably. Administrative statistics provide a useful view of reported generation and recycling, but these figures may not fully describe unreported flows or the activities of informal collection and processing networks. It is therefore important to interpret reported figures carefully while examining the systems intended to improve responsible management.

B. Objectives

The objectives of this study are:

- to examine reported e-waste generation and recycling trends in India;

- to describe the key features of the E-Waste (Management) Rules, 2022;
- to identify environmental, operational, and information-related challenges;
- to discuss practical measures for improving collection, traceability, and circular use of electronic products.

C. Research Questions

The study addresses four questions:

1) How did reported e-waste generation and recycling change between 2020–21 and 2024–25?
2) What role does the EPR framework play in India's e-waste management system?
3) What limitations should be considered when interpreting administrative e-waste statistics?
4) Which actions could improve the safety and effectiveness of e-waste management?

III. Methodology

This paper uses a descriptive secondary-data approach. No new survey, interview, laboratory test, or field observation was conducted. The numerical series is taken from published government/parliamentary reporting summarized in a policy analysis by PRS Legislative Research [6]. The regulatory discussion is based on the official E-Waste (Management) Rules, 2022, government explanations, and the CPCB's E-Waste Management System portal [2]–[4]. The international context is drawn from the Global E-waste Monitor 2024 [1].

The study compares annual reported generation and recycling quantities for five financial years. For each year, a descriptive ratio is calculated as:

$$R_t = \frac{C_t}{G_t} \times 100, \tag{1}$$

where R_t is the reported recycling-to-generation ratio in year t, C_t is the reported quantity recycled, and G_t is the reported generation quantity. Values are rounded to one decimal place for presentation.

This ratio is used only as a descriptive comparison of the two published series. It is not presented as a verified national recycling rate for every item discarded during the year, because the data may not capture all informal or unreported flows and the generation estimate is based on defined reporting methods. The study does not infer causation from changes in the series.

IV. Reported E-Waste Data in India

Table I presents the reported quantities for 2020–21 to 2024–25. The figures show that reported generation increased from 1.35 million tonnes in 2020–21 to approximately 1.61 million tonnes in 2022–23, then decreased in 2023–24 before rising again in 2024–25. Reported recycling increased in every year in the presented series.

TABLE I
Reported E-Waste Data in India

Year	Generated (tonnes)	Recycled (tonnes)	Ratio (%)
2020–21	1,346,496	354,541	26.3
2021–22	1,601,155	527,132	32.9
2022–23	1,609,117	539,256	33.5
2023–24	1,254,287	778,205	62.1
2024–25	1,397,956	1,159,228	82.9

Source: PRS Legislative Research's analysis of parliamentary data [6]. Ratios calculated from displayed values. They are descriptive, not independently verified estimates of all national e-waste flows.

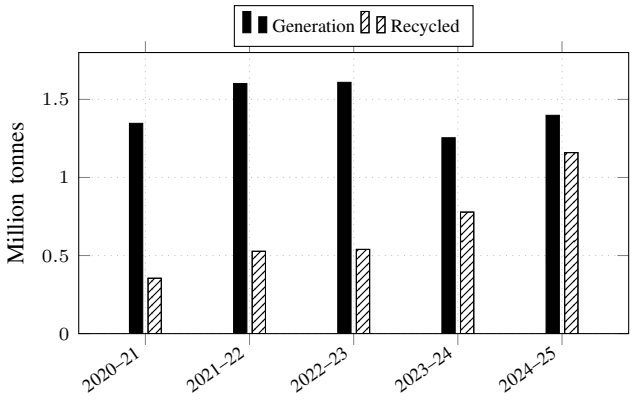


Fig. 1. Reported generation and recycling quantities, converted to million tonnes. Source data: [6].

A. Generation Trend

The reported generation series does not rise uniformly. It increases between 2020–21 and 2022–23, falls in 2023–24, and increases in 2024–25. The drop in the reported figure for 2023–24 should not automatically be interpreted as proof that fewer electronic products were discarded in all parts of the country. Changes in the estimation method, scope, coverage, or reporting context can affect administrative series. Further investigation of the underlying source data would be required before assigning a cause to the year-to-year variation.

B. Recycling Trend

Reported recycling increased from 354,541 tonnes in 2020–21 to 1,159,228 tonnes in 2024–25. This is an increase of approximately 227% relative to the 2020–21 reported quantity, calculated from the two endpoints. The reported recycling-to-generation ratio also increased from 26.3% to 82.9%. This pattern is consistent with a substantial rise in the quantity recorded as recycled in the published series. It does not, by itself, establish that 82.9% of every discarded electronic product generated across India was collected and recycled during 2024–25.

V. Regulatory Framework in India

A. E-Waste (Management) Rules, 2022

The E-Waste (Management) Rules, 2022 were notified under the Environment (Protection) Act, 1986, and came into

force on 1 April 2023 [2], [3]. They superseded the 2016 rules and introduced an updated EPR regime for the equipment categories covered by the rules. The rules have also been amended; researchers and practitioners should consult the current official version rather than rely only on the original notification.

B. Extended Producer Responsibility

EPR places defined end-of-life management obligations on producers of covered electrical and electronic equipment. Under the 2022 framework, registered producers meet applicable recycling obligations through the EPR system, including certificates associated with recycling carried out by registered recyclers. This approach is intended to connect producer obligations with documented recycling activity and to improve accountability across the formal system [2], [5].

EPR can create incentives for producers to support collection and recycling infrastructure. Its effectiveness nevertheless depends on credible documentation, accurate material-flow records, reliable verification, and effective action when requirements are not met. A digital certificate or registration is an administrative mechanism; its environmental value depends on whether the underlying recycling activity is genuine, traceable, and carried out safely.

C. Institutional Roles

The CPCB operates the centralized e-waste EPR portal and provides related guidance and notices [4]. State Pollution Control Boards and Pollution Control Committees have implementation and monitoring responsibilities under the regulatory framework. Producers, manufacturers, refurbishers, and recyclers must meet the requirements applicable to their roles. Bulk consumers and ordinary consumers also influence the system through the way they store, return, sell, or discard equipment.

VI. Environmental and Operational Challenges

A. Hazardous Components and Unsafe Processing

Electronic equipment can contain materials that require controlled handling. Risks depend on the product and the process used. Breaking equipment without suitable controls, open burning of insulated wires, and unsafe treatment of components can release pollutants or expose workers to hazardous substances. Appropriate dismantling, ventilation, protective equipment, pollution controls, and safe residue management are therefore important parts of environmentally sound recycling.

B. Collection and Consumer Participation

A product can only be recycled through an appropriate facility if it enters a collection and return channel. Consumers may retain unused phones and computers at home, sell devices to informal buyers, or discard small electronics with mixed waste. Collection arrangements should be convenient, trustworthy, and clearly communicated. Take-back schemes, collection drives, and authorized drop-off points can reduce friction for consumers, although their performance should be assessed using documented collection data rather than campaign counts alone.

C. Informal Collection and Recycling

Informal actors may collect, sort, repair, dismantle, or trade discarded equipment. These networks can be important for material recovery, but some processing methods may lack occupational safeguards and pollution controls. A practical policy response should distinguish between collection livelihoods and hazardous processing practices. Where feasible, safer collection networks can be connected to registered recyclers through training, transparent purchasing arrangements, and clear rules for handling hazardous fractions.

D. Data Coverage and Comparability

A major limitation in e-waste analysis is that different sources can use different definitions and estimation methods. For example, the Global E-waste Monitor estimates worldwide and national e-waste generation using a harmonized international methodology, while India's CPCB administrative series has been based on defined equipment categories and estimation procedures [1], [3]. As a result, the approximately 4.1 million tonnes reported for India in the international monitor for 2022 should not be compared directly with the 1.61 million tonnes in India's administrative series for 2022–23 as though one disproves the other. They represent different measurement frameworks.

This distinction matters for policy evaluation. A low administrative figure may reflect its defined scope rather than the complete amount of equipment reaching end of life. Similarly, a high ratio of recorded recycling to recorded generation is not necessarily a complete measure of collection coverage if the numerator and denominator do not cover all the same material flows. Public reporting would be stronger if definitions, scope, revisions, uncertainty, and data coverage were made explicit.

VII. Discussion

The secondary data show a substantial increase in the quantity reported as recycled between 2020–21 and 2024–25. This is an important administrative trend, but it should be interpreted alongside the limits of the generation estimate and the distinction between documented recycling and all real-world waste flows. The variation in generation across the years also illustrates why a single year's figure should not be used alone to judge policy performance.

The EPR framework provides a mechanism for assigning obligations and linking them to registered recycling activity. Yet policy effectiveness should be evaluated through multiple indicators: quantities collected, quantities actually processed, recovery of materials, environmentally sound treatment of residues, compliance, and worker safety. Counting registered entities or issued certificates alone does not measure every environmental outcome.

The most effective approach is likely to combine downstream management with upstream prevention. Recycling is essential for equipment that has reached end of life, but repair,

refurbishment, durable design, software support, and reuse can delay disposal. These measures support circular-economy goals by retaining product value and reducing the need for new materials. Their contribution should be measured carefully so that reuse is safe and does not merely transfer unusable equipment to places without adequate treatment capacity.

VIII. Recommendations

Based on the reviewed sources and analysis, the following actions are recommended:

1) **Improve collection access:** Expand convenient authorized return options through retailers, institutions, local authorities, and producer take-back programmes.
2) **Increase data transparency:** Publish annual datasets with clear definitions, scope, estimation methods, revisions, and separate figures for collection, refurbishment, recycling, and residue disposal.
3) **Strengthen traceability:** Audit material flows and supporting records so that recycling claims correspond to actual processing at registered facilities.
4) **Support safer livelihoods:** Provide training and practical pathways for informal collectors to connect with formal collection and recycling systems, while discouraging unsafe dismantling and burning.
5) **Promote repair and reuse:** Encourage repair services, refurbishment, spare-parts availability, and responsible reuse before products become waste.
6) **Build consumer awareness:** Explain where to return old devices and why batteries and electronic components should not be handled as ordinary mixed waste.
7) **Evaluate environmental outcomes:** Monitor not only tonnes reported as recycled, but also pollution controls, residue management, occupational safeguards, and recovery efficiency.

IX. Limitations and Future Work

This paper is limited by its reliance on published secondary sources and the availability of comparable national data. It does not independently verify the reported quantities, inspect recycling facilities, or measure unreported informal activity. The calculated recycling-to-generation ratios are descriptive and inherit the limitations of the source figures. The paper also does not estimate the financial cost of interventions or compare the environmental performance of individual recycling technologies.

Future work could use a consistent multi-year dataset with transparent definitions, conduct interviews with authorized recyclers and collection workers, survey consumer disposal behaviour, and compare state-level collection infrastructure. A separate study could examine whether public take-back programmes improve the proportion of discarded equipment reaching registered facilities. Any such study should document its sampling method and distinguish measured results from assumptions.

X. Conclusion

E-waste management in India involves environmental protection, resource recovery, public participation, and the regulation of complex collection and recycling networks. The reviewed national series reports an increase in recycling quantities from 354,541 tonnes in 2020–21 to 1,159,228 tonnes in 2024–25, while reported generation varies across the same period. These figures suggest a significant change in recorded recycling activity, but they must be interpreted in light of differences in scope, estimation methods, and potentially unreported waste flows.

The E-Waste (Management) Rules, 2022, and their EPR framework provide an important structure for producer obligations and registered recycling. Continued progress requires more than regulation alone: convenient collection, reliable traceability, transparent data, safe treatment, and support for repair and reuse are also essential. A circular-economy approach that extends product life and ensures responsible recycling can help India manage electronic waste more sustainably.

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